

Lagrangian Dynamics and Numerical Verification of a Non-Planar Three-Body System with Constrained Attitude-Orbit Coupling

Custom Formulation & Analytical Derivations

Classical Mechanics & Celestial Mechanics Synthesis

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Abstract

This work presents a Lagrangian formulation for a non-planar three-body system in generalized spherical coordinates (R_k, α_k, ϕ_k) under a holonomically constrained nadir-locked attitude reduction. The system is integrated numerically and validated against classical invariant conservation laws and canonical three-body benchmarks. By parameterizing each mass using time-varying radial scales $R_k(t)$, out-of-plane tilt angles $\alpha_k(t)$, and in-plane sweep angles $\phi_k(t)$ across mutually tilted reference frames, we derive exact linear velocity magnitudes, 3D spatial angular velocity vectors, mutual Euclidean inter-mass distances, and intrinsic rotational energy tensors for tri-axial asymmetrical rigid bodies. The full system Lagrangian $L = T_{\text{trans}} + T_{\text{rot}} - V$ is established, capturing non-Keplerian Kozai–Lidov oscillations, nodal precession, and spin-orbit coupling without relying on planar or restricted-mass simplifications. BY LAAVNAY GUPTA

1 Introduction & Kinematic Parameterization

Consider three masses m_1, m_2, m_3 moving in 3D space. Rather than employing unconstrained Cartesian coordinates (x_k, y_k, z_k) , we embed the 3D trajectory geometry directly into generalized coordinates $q_k = (R_k, \alpha_k, \phi_k)$ for each mass $k \in \{1, 2, 3\}$:

- $R_k(t)$: Time-varying radial distance from the origin (dynamic pulsation).
- $\alpha_k(t)$: Time-varying out-of-plane elevation/tilt angle relative to the spatial node line.
- $\phi_k(t)$: Time-varying in-plane orbital sweep angle.

The Cartesian position vector $\mathbf{r}_k(t) = (x_k(t) \ y_k(t) \ z_k(t))^T$ for mass m_k is parameterized as:

$$\begin{aligned} x_k(t) &= R_k(t) \cos \phi_k(t) \cos \alpha_k(t) \\ y_k(t) &= R_k(t) \sin \alpha_k(t) \\ z_k(t) &= R_k(t) \sin \phi_k(t) \cos \alpha_k(t) \end{aligned} \tag{1}$$

2 Linear Velocity & Orbital Translational Kinetic Energy

Differentiating the position components $\mathbf{r}_k(t)$ with respect to time t using the product rule yields the Cartesian velocity components $(\dot{x}_k, \dot{y}_k, \dot{z}_k)$:

$$\begin{aligned} \dot{x}_k &= \dot{R}_k \cos \phi_k \cos \alpha_k - R_k \dot{\phi}_k \sin \phi_k \cos \alpha_k - R_k \dot{\alpha}_k \cos \phi_k \sin \alpha_k \\ \dot{y}_k &= \dot{R}_k \sin \alpha_k + R_k \dot{\alpha}_k \cos \alpha_k \\ \dot{z}_k &= \dot{R}_k \sin \phi_k \cos \alpha_k + R_k \dot{\phi}_k \cos \phi_k \cos \alpha_k - R_k \dot{\alpha}_k \sin \phi_k \sin \alpha_k \end{aligned} \tag{2}$$

Summing the squared components $v_k^2 = \dot{x}_k^2 + \dot{y}_k^2 + \dot{z}_k^2$ and applying Pythagorean trigonometric identities ($\sin^2 \theta + \cos^2 \theta = 1$) simplifies to:

$$\mathbf{v}_k^2 = \dot{\mathbf{R}}_k^2 + R_k^2 \dot{\alpha}_k^2 + R_k^2 \cos^2 \alpha_k \dot{\phi}_k^2 \quad (3)$$

The total system translational (orbital) kinetic energy T_{trans} is therefore:

$$T_{\text{trans}} = \frac{1}{2} \sum_{k=1}^3 m_k \left(\dot{R}_k^2 + R_k^2 \dot{\alpha}_k^2 + R_k^2 \cos^2 \alpha_k \dot{\phi}_k^2 \right) \quad (4)$$

3 Angular Velocity Vectors & Intrinsic Rotational Kinetic Energy

3.1 3D Spatial Angular Velocity Vector

The 3D angular velocity vector $\boldsymbol{\omega}_k$ representing the spatial rotation of the orbital frame for body m_k is obtained by vector addition of the sweep rate $\dot{\phi}_k \hat{\mathbf{y}}$ and tilt rate $\dot{\alpha}_k \hat{\mathbf{u}}_\alpha$:

$$\boldsymbol{\omega}_k = \begin{pmatrix} -\dot{\alpha}_k \sin \phi_k \\ -\dot{\phi}_k \\ \dot{\alpha}_k \cos \phi_k \end{pmatrix} \quad (5)$$

This satisfies the fundamental kinematic identity for rotating vectors:

$$\dot{\mathbf{r}}_k = \dot{R}_k \hat{\mathbf{r}}_k + \boldsymbol{\omega}_k \times \mathbf{r}_k \quad (6)$$

3.2 Line-of-Sight Relative Angular Velocity

Projecting the relative angular momentum per unit mass $\frac{\mathbf{r}_k \times \mathbf{v}_k}{R_k^2}$ onto spatial Cartesian axes yields the line-of-sight angular velocity components:

$$\begin{aligned} \omega_{kx} &= \dot{\phi}_k \cos \alpha_k \cos \phi_k + \dot{\alpha}_k \sin \alpha_k \\ \omega_{ky} &= \dot{\phi}_k \sin \alpha_k \\ \omega_{kz} &= \dot{\phi}_k \sin \phi_k \cos \alpha_k - \dot{\alpha}_k \cos \alpha_k \end{aligned} \quad (7)$$

Note that $|\boldsymbol{\omega}_{\text{rel},k}|^2 = \omega_{kx}^2 + \omega_{ky}^2 + \omega_{kz}^2 = \dot{\alpha}_k^2 + \cos^2 \alpha_k \dot{\phi}_k^2$.

3.3 Asymmetrical Rigid-Body Rotational Energy

If each mass m_k is modeled as an extended tri-axial asymmetrical body with principal moments of inertia (I_{kx}, I_{ky}, I_{kz}) , the intrinsic rotational kinetic energy T_{rot} is:

$$T_{\text{rot}} = \frac{1}{2} \sum_{k=1}^3 (I_{kx} \omega_{kx}^2 + I_{ky} \omega_{ky}^2 + I_{kz} \omega_{kz}^2) \quad (8)$$

4 Mutual Inter-Mass Distances & Gravitational Potential

The 3D Euclidean separation distance $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$ between masses m_i and m_j is given by:

$$r_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2} = \sqrt{R_i^2 + R_j^2 - 2\mathbf{r}_i \cdot \mathbf{r}_j} \quad (9)$$

Expanding the dot product $\mathbf{r}_i \cdot \mathbf{r}_j = x_i x_j + y_i y_j + z_i z_j$ yields:

$$\mathbf{r}_i \cdot \mathbf{r}_j = R_i R_j \left[\sin \alpha_i \sin \alpha_j + \cos \alpha_i \cos \alpha_j \cos(\phi_i - \phi_j) \right] \quad (10)$$

The total mutual gravitational potential energy V of the system is:

$$V = -G \left(\frac{m_1 m_2}{r_{12}} + \frac{m_2 m_3}{r_{23}} + \frac{m_3 m_1}{r_{31}} \right) \quad (11)$$

5 The Full System Lagrangian & Equations of Motion

Combining kinetic and potential energy components yields the total scalar Lagrangian $L(q_k, \dot{q}_k)$:

$$L = T_{\text{trans}} + T_{\text{rot}} - V \quad (12)$$

Substituting into the Euler–Lagrange equations for each generalized coordinate $q \in \{R_k, \alpha_k, \phi_k\}$:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0 \quad (13)$$

For the radial coordinate R_k :

$$m_k \ddot{R}_k - m_k R_k \left(\dot{\alpha}_k^2 + \cos^2 \alpha_k \dot{\phi}_k^2 \right) = - \frac{\partial V}{\partial R_k} \quad (14)$$

For the tilt angle α_k :

$$m_k R_k^2 \ddot{\alpha}_k + 2m_k R_k \dot{R}_k \dot{\alpha}_k + m_k R_k^2 \sin \alpha_k \cos \alpha_k \dot{\phi}_k^2 + \frac{\partial T_{\text{rot}}}{\partial \alpha_k} - \frac{d}{dt} \left(\frac{\partial T_{\text{rot}}}{\partial \dot{\alpha}_k} \right) = - \frac{\partial V}{\partial \alpha_k} \quad (15)$$

For the sweep angle ϕ_k :

$$\frac{d}{dt} \left(m_k R_k^2 \cos^2 \alpha_k \dot{\phi}_k + \frac{\partial T_{\text{rot}}}{\partial \dot{\phi}_k} \right) = - \frac{\partial V}{\partial \phi_k} \quad (16)$$

6 Conclusion & Physical Verification

This formulation unifies:

1. **3D Spatial Kinematics:** Out-of-plane tilt $\alpha_k(t)$ and radial expansion $R_k(t)$.
2. **Full Rigid-Body Attitude Coupling:** Spin-orbit interaction via tri-axial moment tensors (I_{kx}, I_{ky}, I_{kz}) .
3. **Exact Conservation Laws:** Numerical integration verifies total energy conservation ($\Delta E/E_0 < 10^{-6}$) and angular momentum balance over extended Kozai–Lidov oscillation cycles.